Haesung Pyun

Ph.D. Student | Language Model Reasoning, Tool Use, and Evaluation
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EDUCATION

Seoul National University Sep 2023 - Aug 2028 (expected)
Ph.D. in Data Science, Graduate School of Data Science
Advisor: Yohan Jo

Hanyang University 2017 - 2023
B.S. in Electronic Engineering, Magna Cum Laude

RESEARCH EXPERIENCE

Seoul National University, Graduate School of Data Science 2023 - Present
Ph.D. Student | Advisor: Yohan Jo

- **PA-Tool (ACL 2026):** Contributed to training-free tool-schema adaptation for small LMs. Tested the role of pretrained knowledge through supervised fine-tuning (SFT) experiments; the method reduced schema-misalignment errors by 80%.
- **KAPPA (ICML 2026, co-first):** Co-formalized the gap between information encoded in hidden representations and predicted answers. Implemented an inference-time intervention aligning knowledge and prediction subspaces on multiple-choice questions.
- **CombiSearch (ACL 2025, first author):** Developed combinatorial in-context example selection and analyzed oracle-versus-retriever gaps in dialogue state tracking. Retrievers trained with CombiSearch surpassed baselines on MultiWOZ using 5% of their training data.
- **SHAPE (ICML 2026 Workshop):** Co-authored research analyzing mathematical reasoning trajectories and how RL with verifiable rewards (RLVR) changes the range of solution strategies.

Seoul National University Jan 2023 - Aug 2023
Research Intern, Structure & Knowledge Injection into Machine Learning Lab | Advisor: Jay-Yoon Lee

- **SEAL-SSL:** Implemented an energy-based objective for semi-supervised learning and structured knowledge injection, using knowledge distillation, REINFORCE, and task adaptation.

Seoul National University, Machine Intelligence Laboratory Mar 2022 - Dec 2022
Research Intern | Advisor: Kyomin Jung

- Developed a multi-view, domain-agnostic intent-induction method using K-means and multitask learning; published at the DSTC11 Workshop.

Korea Institute of Science and Technology, AI Research Center Dec 2021 - Feb 2022
Research Intern | Advisors: Heeseung Choi and Juyoun Park

- Built a tourism recommendation demo with matrix factorization and higher-order connectivity; evaluated recommendations through A/B testing and user feedback.

SELECTED PUBLICATIONS

- Y. Park*, **H. Pyun***, Y. Jo. *Bridging the Knowledge-Prediction Gap in LLMs on Multiple-Choice Questions.* **ICML 2026.**
  ** Equal contribution*
- J. Lee, W. Song, J. Han, **H. Pyun**, Y. Jo. *Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models.* **ACL 2026.**
- **H. Pyun**, Y. Park, Y. Jo. *Improving Dialogue State Tracking through Combinatorial Search for In-Context Examples.* **ACL 2025.**
- J. Song, S. Song, M. Oh, **H. Pyun**, S. Lee, Y. Jo. *SHAPE of Chain-of-Thought in Math Reasoning.* **AI for Math Workshop, ICML 2026.**

HONORS TEACHING AND SERVICE

- **NRF Fellowship:** National Research Foundation of Korea, 2-year graduate research fellowship, 2025 - 2027.
- **Teaching:** Computing for Data Science 1 (Spring 2026); Conversational AI (Fall 2024), SNU GSDS.
- **Service:** Reviewer, Pluralistic Alignment Workshop at ICML 2026.